# DIPOLOG BAMBOO RONDALLA

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**Grade Level:** 10
**Area:** Music

## I. OBJECTIVES

### A. Content Standards

Demonstrate an appreciation and general understanding of rondalla music:
- history, origin of the instruments, the ensemble
- different instruments and their roles

### B. Performance Standards

1. Follow specific instrument lines in a rondalla piece and sing the lines aided with sheet music (if available).

2. Perform in groups rondalla pieces (with or without instruments).

3. Arrange and perform in groups rondalla-style renditions of songs of their own choice or invention.

4. Demonstrate sensitivity to demands of ensemble performance (volume swells, dynamics, etc.).

5. Analyze complex issues (even non-musical ones) with the help of an analytical framework.

### C. Learning Competencies/Objectives

1. Physical/Sensory: Make physical (as much as possible) contact with rondalla music and instruments. Experience playing or, at least, imagining playing). Appreciate the power of technologies such as music theory, instrument construction, arranging and conducting, etc.

2. Historical/Psychological: Have an idea of how rondalla became part of the Philippine musical ecosystem. Be aware of one's gut response to the ensemble's music and instruments.

3. Mythical/Symbolic: Have a sense of the potential power of the ensemble as a cultural element. Have a sense of how traditions can change through innovation.

4. Integrative/Unitive: Experience the flow state of ensemble playing. Experience being part of a larger body.

## II. CONTENT

A. Rondalla: Tradition, Innovation, Emotion
B. An integrative experience and exploration of the rondalla ensemble

## III. LEARNING RESOURCES

### A. References

1. Print Materials

   a. BMI ethnography of Dipolog's Bamboo Rondalla
   b. Schematics of both bamboo and non-bamboo rondalla instruments (for comparison)

2. Audio-Video Materials

   a. Videos of Dipolog's Bamboo Rondalla from BMI documentation: *http://tiny.cc/BMI-Dipolog_Rondalla*
   b. Videos of Jay Sarita's workshop from BMI documentation: *http://tiny.cc/BMI-Sarita-Workshop*

3. Music Scores
   Sheet music of rondalla arrangements featured in the videos

### B. Other Learning Materials

1. Actual rondalla instruments, pictures, schematic diagrams, cross-sections.

2. Illustration of integrative analytical framework (suggested: Ken Wilber's AQAL - All Quadrants, All Levels. This is not necessarily for student use, but more for the teacher. This framework, among others, can be applied to any level and any subject.)

## IV. PROCEDURE

Progress from objective contact with the material towards some form of understanding and application of the material. Given focused drilling, mastery may emerge. Technical mastery of the rondalla in a school year quarter is NOT the point of this module.

### A. Reviewing the previous lesson or presenting the new lesson

1. Begin with experiencing rondalla music. Allow students to close their eyes and/or move-dance and/or draw – whatever the music makes them feel like doing.

2. Teacher draws out reactions and synthesizes the experience. (Suggested framework is Edward de Bono's Six Hats style of idea and discussion management. Cluster ideas into these 6 "hats": Facts or verifiable quantities, Feelings or strong subjective reactions, Optimism or the good things that may arise, Pessimism or the bad things that may arise, Alternatives or other options, Executive how to decide and what action to take.

## B. Establishing the purpose of the lesson

1. The big takeaway here is to have a realistic view of culture and how it evolves using the rondalla as a case study.
2. The Rondalla Ensemble is an icon of Philippine music. Although its distinctive timbre is hardly heard in contemporary sound recordings, the tradition is alive and well in the context of schools, celebrations and public events that require some nod to the concept of Filipiniana.
3. What is its value to you as a listener? As a musician? Is it worthy of a significant investment of your time and effort? Is there joy in it? *Makakain ba yan?!* What is its value on the global music stage? Why is it not on the MTV Channel? Should it be?
4. Ideally the module may deepen to life-skill discussion such as:
   a. How culture and technology are intertwined (For a diagram, see "Siningbayan: The Art of Nation-Building:" *www.blafi.org*).
   b. Tradition, obsolescence, and survival of the most competent
   c. Creating the future you want to see – like Jay Sarita, the bamboo rondalla instrument-maker.

## C. Presenting examples/instances of the new lesson

1. Compare the sounds of traditional wooden rondalla instruments to those made of bamboo.
2. Try miming one short piece of each and feel the differences and similarities.
3. The teacher may use Six Hats to manage discussions - drilling students in this and other analytical frameworks sharpens and speeds up analytical processes. This is an asset across all fields of human endeavor!

## D. Discussing new concepts and practicing new skills # 1

1. Imagining an activity stimulates the same neural and muscular pathways as the actual activity similar to visualization exercises done by top coaches and athletes.
2. Subjective oriented approach: Learners can practice a rondalla piece even without instruments by miming a piece - playing along and acting out their part in the ensemble. This may be funny at first and that's good and alright AS LONG AS the chosen piece is rehearsed (repeated over and over again until it feels presentable).
3. NOTE: Such an exercise (you can make up others) will give the experience of drilling-to-mastery even without the actual instruments and time to "burn in" the micro-muscle memory. These subjectively oriented exercises may be applied to non-musical life situations such as public speaking and going on dates. Imagining the situations handled with confidence will lead to actual confidence in actual situations.

## E. Discussing new concepts and practicing new skills # 2

1. Objective oriented approach: Using videos of Jay Sarita's workshop, have the students inspect the physicality of the instruments. Look at the diagrams. Discuss the science and math involved in instruments and music.
2. Draw out and cluster ideas, deepen discussion to how Jay may be reshaping a tradition and how this relates to young people who are still deciding what to do with their lives. Is it safer to follow what others have done before you? What are the risks of innovation? (Advanced but probably inevitable discussion: Will technology make human beings obsolete?)

## F. Developing mastery (leads to formative assessment)

Assumption if instruments are available: General guideline for all Mastery activities such as memorizing and rehearsing: The saying practice makes perfect is only true if you practice perfectly. This usually involves practicing slow and perfect progressing towards fast and perfect.

## G. Finding practical applications of concept and skills in daily living

Prompt 1: Remember an inharmonious or problematic situation - perhaps an argument or confrontation. If the situation were a piece of music, how would you play your part in order to make the music sound better?

Prompt 2: When you are talking with someone be aware of your reactions to the other. Are you playing harmony or counterpoint? Are you speeding up or slowing down?

## H. Making generalization and abstraction about the lesson

Sort out information, observations, feelings, criticism, etc., using this integrative quadrant by Ken Wilber, AQAL. (For a diagram, see "Siningbayan: The Art of Nation-Building:" *www.blafi.org*):

- Upper Right quadrant (UR): The **Individual Objective** quadrant. It holds details of objects - instruments, plectrums, strings, construction, materials, tools, costs, etc.
- Lower Right quadrant (LR): The **Collective Objective** quadrant. It holds details related to systems - the rondalla ensemble, the sum total of parts that add up to a complete instrument, the training of arrangers and players, the historical events that made the rondalla a Filipino orchestra, etc.
- Upper Left quadrant (UL): The **Individual Subjective** quadrant. Your feelings about rondalla instruments and music, any ideas, suggestions, desires and criticisms. Personal reactions or points of inspiration sparked by the materials, etc. May include critical appreciation of rondalla performances, arrangments, and musicians' skills and techniques.
- Lower Left quadrant (LL): The **Collective Subjective** quadrant. What role does rondalla play in Philippine arts and culture? Do people in general identify with the ensemble? How can rondalla strengthen your nation?

### I. Additional activities for application or remediation

Work with real instruments for exploration, discovery, and eventual mastery.

## V. TEACHER'S REMARKS

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## VI. REFLECTION

A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
C. Which of my teaching strategies worked well? Why did this work?
D. What difficulties can my principal or supervisor help me solve?
E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

**Excerpt from the BMI Ethnography of Dipolog's Bamboo Rondalla**
*"Bamboo Music Instruments"*
*By Sol Maris T. Trinidad*

USING BAMBOO

When Jay Sarita first ventured into bamboo xylophone-making, the reading materials he could find were only about making wood xylophones. Applying some of the techniques for wood, he started making his own bamboo xylophones. The first batch he made was an admitted failure. It was so bad that he wanted to just scrap the idea. However, he continued striving and found that the secret to using bamboo as a material was on the bamboo itself. He realized that as a material, it has very different characteristics from wood. To date, he still has not encountered a resource material for bamboo.

During the Tunogtugan Festival, Jay met Dr. Wu Shih-Yin and his Taiwan Bamboo Orchestra (TBO). Dr. Wu is a scientist who specializes in using bamboo as raw material for the instruments of the TBO. Jay learned many techniques on making bamboo instruments like the xylophone from Dr. Wu. When DCR went to perform in the Taiwan leg of the 5th International Rondalla Festival, Jay grabbed the opportunity to visit Dr. Wu's office. He spent three meetings with Dr. Wu in the latter's work laboratory, studying everything that he could in that short amount of time.

Jay also collaborated with the guitar maker, Adolfo Timuat. They discussed at length on the various characteristics of wood and acoustics. With Mr. Timuat, Jay learned all that he could about guitar-making. One of Jay's biggest takeaways was that no matter how meticulously one works with wood, the real outcome is not always what one reads; not all processes for wood are applicable to bamboo. He says that nothing beats actually working with the material and being open to asking questions and getting answers. The products that he produces are a result of his own experimentation.

When working with bamboo, Jay has to take the nodal points of the bamboo into consideration: the closer to the node, the denser or thicker the material. This makes its flexibility incomparable to wood, i.e., a 2-inch-by-4-inch-sized wood would have an even grain. With a bar of wood, the left and right ends are very similar, almost uniform. With bamboo, it is very different and looks like it is all over the place. Some bamboo is also crooked both laterally and longitudinally. It is difficult working with bamboo.

As of writing, *buntong or botong (Dendrocalamus latiflorus)* is the ideal material for Jay's instruments. Other varieties of bamboo available in Dipolog like the *lunas* (*Bambusa nguyenii Ohrnb.*) and *tungkan (Bambusa blumeana)* proved to be too thin for his needs. Buntong or botong can be found eight kilometers away from Dipolog City, both in upland and low-lying areas. The farther from the city, the more of this variety may be found, as long as the area is near a river. Jay has a regular contact person from whom he sources this bamboo. He entrusts the choice of quality bamboo to this person who is of advanced age. He also likes that he purchases them at a reasonable price.

The schedule for harvesting the bamboo is left to his contact as some of the considerations are the shifting of the tides and the like. When ordering bamboo, he has a waiting time of one week. But Jay does not use this batch of bamboo yet. The stock that he used recently in making his prototype were bought just after the 2013 Tunogtugan

Festival. He air-dried and stored these 12 pieces of 8 feet long poles, upright, on a concrete flooring in his backyard. He made sure to treat his stocks with termite spray, Solignum. He has used a lot of this bamboo batch from 2013.

Save for naturally air-drying the bamboo for long periods, he has yet to use other drying techniques like smoking. Thus far, he has only purchased bamboo twice: in 2013 and most recently in 2019. He avoids keeping too much stock as it might decay or be infested and become unusable.

Jay can make approximately 10 units or more of a 3-octave chromatic xylophone from 12 bamboo poles. While the bars are made from botong, the frame is made from other thinner bamboo varieties like tungkan and other local ones. When Jay would need a 1x2 cut size, he would do two layers using the botong but up to five layers of the other bamboo types.

Before, Jay used bamboo only for the bars. The resonators were usually made of polyvinyl chloride (PVC) pipes as PVC is easy to work with. Recently, however, Jay and his workers were able to process bamboo into square-shaped cylinders as resonators. At first, this was difficult because when they cut the bamboo with a bolo, it did not come out straight like wood; later, they discovered that the safest and straightest way to cut the bamboo into square resonators is to handsaw it.

To make bamboo resonators, they first draw a line on one side and a parallel one on the other side using a ruler and a pen. These lines serve as their guide for sawing the bamboo with a bow-type hand saw. To get the desired width, they process the cut bamboo using a thickness sander. With this machine, Jay can dictate how thin the bamboo will be, even up to veneer size. All the instruments made in their shop go through this sander. This sander was made by guitar markers. He bought it from them for Php15,000 plus shipping fee of approximately Php 3,000.

The decision to change the resonators to square from cylinder was made due to the difficulty in sourcing the appropriate bamboo dimensions to match the thickness of the xylophone bars. This also gave him the upper hand in presenting that they have the capacity to make resonators from both PVC and bamboo, cylindrical and square. He also saw that the bamboo xylophones available in the market only had the frames made of bamboo with the bars made from narra and the cylinders from PVC. So, when he says that they have bamboo xylophones, he can rightfully claim that everything about it is made from bamboo.

One of their products is the table-top xylophone. This is unique in that the resonator is the travelling case of the instrument. He designed it because it made the xylophone easier to carry. It can even be hand-carried into an airplane due to its compact body. He started selling this in 2017.

Prof. Jocelyn Guadalupe shared during the interview that Atty. Dulce Punsalan was able to show her the table-top xylophone prototype from Jay. The attorney found the instrument to be very durable between various climate conditions as she had brought it with her in international trips. The tuning did not also change. (Prof. Guadalupe said that the drawback with bamboo was that the tuning can change depending on the temperature in the area.) Jay attributed this durability to the botong bamboo material.

BAMBOO RONDALLA

The *rondalla* instruments that Jay has made from bamboo are comprised of the *5 bandurria, 2 octavina, 2 laud*, 2 classical guitars, 1 bass, and 1 marimba. He differentiated between the regular bandurria and what he called the RS model (named after National Artist for Music Ramon P. Santos) of which he made one. The octavinas differ from the regular guitar-shaped octavina in that it is tear-dropped shaped. The design of the laud also had a different approach. It took Jay and his workers three months to make all 13 instruments.

The length of aging process for the bamboo used in this particular set of rondalla instruments is the same as that of the xylophone. The bamboo also came from the same clump of bamboos. However, the xylophone made use of a much older bamboo. As the bamboo they used was already available, the processing time did not include the bamboo preparation time.

Making of a bamboo *rondalla* was a standing challenge from Ms. Aurea Lopez after she had visited his workshop in 2014. She asked if it was possible to make a *bandurria* from bamboo. Because Jay answered that it was possible, she then asked him to make one. The idea, however, was shelved and forgotten for five years. Later, he got married and became less active in the *rondalla* performing scene. One day, he asked one of his workers about making a *bandurria* from bamboo. His worker answered that it was easy to make one and that they could start building it the next day. So, Jay, with four of his workers, started studying the mechanics of making the instrument.

One of the questions they had was on the ideal process of binding the bamboo. At that time, he had no idea about veneers. When thin enough, bamboo can be torn by hand. What Jay found out was that companies like Yamaha or those in China used veneer for their guitars but arranged them cross-grain. They made the observation that the sound of solid wood was louder than instruments made from the cross-grain veneer. This made Jay think of not using the cross-grain method in binding bamboo. There was no available information on working with bamboo, but the ready-made cross grain veneer instruments already produced weak sound boards.

Jay decided to make things differently by doing a same-grain two-layered veneer for their *rondalla* materials: The veneer was cut in approximately 1-inch widths and joined together as the 1st layer. The second layer was also made of the same veneer widths, but the joinery of the 1st layer was in between the joinery of the 2nd layer. In the event that one layer fails, there was still another layer that provided support. This resembled the idea of hollow blocks, alternating or interlocking. After making their first bamboo soundboard, they compared it with a spruce sound board. Bending both laterally produced the same results. Cross-grain veneers could not be bent in the same way. And so, they continued with making the instrument.

The first one they finished was a *bandurria*. However, Jay wanted to test the sound immediately, so even before the finished *bandurria* got varnished, Jay had already attached the strings. He observed that the lower registers sounded just like the *bandurria*s made from wood, but the higher register notes had a thin quality about them that was unsatisfactory. So, they took the instrument apart and further thinned-out the sound board. The new sound board produced an acceptable sound and they continued making the rest of the instruments.

In coming up with his bamboo instruments, Jay did not visit other manufacturers anymore. His collaborations were with other *rondalla* musicians like Ben Brillantes, a luthier from Dumaguete, and William Alama, who was also involved in the Gitara Ni Juan Project of the Department of Science and Technology. The same-grain veneers were entirely his idea that he and his workers put into work. For their instruments, they adopted the coconut shell as material for the finger boards from Mr. Alama. Any ba-o ng niyog can be used, no particular coconut variety was identified. The shell was cut into 1 cm x 1 cm or 15 mm x 15 mm tiles for their use. Jay found the results very beautiful in sound. While he had asked permission from Mr. Alama in also using coconut shells, the process that Jay's workers put it through is different from Mr. Alama's. During a visit at Jay's workshop, Mr. Alama commented that their processing time was shorter, and Jay's workers finished faster.

All the wooden materials used in the instruments of the bamboo *rondalla* are from bamboo except for the neck, which is made from mahogany. The sound board, side, and back are from bamboo. Some of their finger boards are made from the coconut shell while the others are from local wood like mahogany. The bracing of the *bandurria* is also from wood. Even if Jay wanted the finger boards to be made from bamboo, he was concerned that the fibrous quality of the materials will injure the musician. This made him decide to use wood or the coconut shell as finger boards.

For comparison, they used bamboo for the bracing of one of their classical guitars and wood for the other one. For comparison also, one classical guitar used wood for the front and back. Jay observed that the guitar using only bamboo all over was better sounding to his ears than the one that had wood mixed in. Jay surmised that the frequency of bamboo was united in the all-bamboo guitar.

With the laud he made, they changed the position of the f hole and computed the load of the bridge to make it hold the instrument despite the effects of changes in the weather on the instrument. Jay also decided to change the design of the octavina to make the *rondalla* family of instruments the same size and visually similar. Despite the new shape of the octavina, Jay states that it is still the same octavina instrument because the scale length is the same. This made Jay interested to adopt the tear-drop shaped bass that Mr. David Dino Guadalupe showed him.

For the players of the instrument, the bamboo instruments are heavier than the wooden ones, especially the bass. However, there have been no complaints from the players so far in terms of action, playability, or even how they hold the instrument or seat with it. Some did not even notice that they were already using the bamboo *rondalla*.

If compared, the sound of his bamboo and wood *rondalla* are close. At the moment, if their concert *bandurria* has a 10 rating, with 1 as the lowest, the bamboo *bandurria* prototype is at 8. In terms of cost of materials, working with bamboo is a whole lot cheaper than working with wood, especially spruce. However, in terms of manpower hours, bamboo instruments cost a lot more. A final costing has not yet been made as of writing.

At the moment, there are no noticeable storage or care requirements as these instruments were just recently finished. Jay is waiting for any of them to fail as this will help in improving the instrument.

Some makers just build the instruments without thinking. Jay's advocacy is to come up with ways to make the instruments more professional-looking with a beautiful sound – the whole package – for as long as the resources are available and accessible. Jay says that if there is something we can contribute to this, then we should. The ultimate purpose is to raise the standard of the *rondalla* instruments.